HACK A CAUSE

"जलमंथन" - BRAINSTORMING ON WATER ISSUES

1.AI-Enabled Smart Sanitation & Access Locator

Problem Statement:

Create an AI-driven system that monitors hygiene conditions of public toilets in real time and displays a live cleanliness grade outside the facility. The platform should also guide users to alternative nearby toilets within 1–2 km, showing their location, rating, and operational status.

Key Problems to Address:

- Computer vision–based cleanliness detection (litter, wet floors, overflow)
- Water availability and usage monitoring
- Automated hygiene scoring algorithm
- Geo-mapping of nearby sanitation facilities
- Smart suggestion of next best option (1–2 km radius) with rating comparison
- Live status dashboard and public display interface

Tech Stack Options:

1. Python + OpenCV + TensorFlow + Firebase + Google Maps API
2. MERN Stack + Vision API + Mapbox
3. Flutter + FastAPI backend + ML model integration + PostgreSQL**

2.Integrated Drought Warning & Smart Tanker Management System

Problem Statement:

Design an integrated digital platform that predicts emerging drought stress using rainfall and groundwater trends and enables district authorities to proactively plan and optimize water tanker allocation. The system should shift response from crisis management to preventive water governance.

Key Problems to Address:

- Rainfall deviation and groundwater trend analysis
- Village-level Water Stress Index generation
- Predictive tanker demand estimation
- Priority-based allocation (population + severity)
- Route optimization for tanker dispatch
- Real-time monitoring and decision dashboard

Tech Stack Options:

1. Python + FastAPI + PostgreSQL + React + Google Maps API
2. MERN Stack + Weather API + Mapbox + Basic optimization algorithm
3. Django + Pandas (data modeling) + Leaflet.js + Firebase

3.Smart Household Tank & Tap Water Quality Monitoring System

Problem Statement:

Develop a low-cost IoT-enabled solution that monitors water quality in household overhead tanks, storage containers, and kitchen taps. The system should detect contamination during storage or distribution within the home and provide real-time alerts to residents.

Key Problems to Address:

- Water contamination due to unclean overhead tanks
- Quality deterioration in stored water (drums, underground tanks, buckets)
- Sudden tap water contamination after irregular supply or pipeline repair
- Detection of abnormal pH, turbidity, and TDS levels
- Stagnation-related quality decline
- Offline/SMS-based alert support for low-connectivity rural areas

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Tech Stack Options:

1. ESP32 simulation + Python (FastAPI) + Firebase + React
2. Python + Pandas + Scikit-learn (anomaly detection) + Streamlit
3. Flutter + Firebase + SMS Gateway API + Threshold-based alert logic